Problems Set 4: Model Selection

Detailed, Self-Explanatory Step-by-Step Solutions

BSDS V: Regression Techniques

Prepared from the supplied Problems Set 4 and the uploaded Regression Techniques lecture-note source, following the structure of the supplied PS1/PS2 solution sets.

How to use this solution set

Each question is presented in the following order:

1. the complete question as it appears in the assignment;
2. the exact model-selection and regression theory needed from the supplied notes;
3. a step-by-step derivation, with each algebraic transition explained;
4. a boxed final result and, where useful, its interpretation.

Notation and standing assumptions

There are K candidate predictors in the full model. A particular subset model M contains p_M predictors, and the full model F contains all K predictors. Every model includes an intercept. Thus model M has $p_M + 1$ fitted coefficients and residual degrees of freedom $n - p_M - 1$; the full model has $K + 1$ coefficients and residual degrees of freedom $n - K - 1$. We use

$$SS_Y = \sum_{i=1}^{n} (Y_i - \bar{Y})^2$$

for the total sum of squares about the mean.

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1 Question 1: Relationship between scaled RSS and adjusted R^2

Question (as assigned)

1. For a model M, let $\overline{R}^2_M$ be the adjusted R^2 and RSS_M be the residual sum of squares under M. Show that

$$\frac{\text{RSS}_M}{\text{RSS}_F} = \frac{(n-K-1)(1-\overline{R}^2_M)}{(n-p_M-1)(1-\overline{R}^2_F)},$$

where p_M is the number of predictors involved in the model M.

Required theory from the supplied lecture notes

The supplied Model Selection notes define the coefficient of determination for a model M by

$$R^2_M = 1 - \frac{\text{RSS}_M}{SS_Y},$$

and the adjusted coefficient of determination by

$$\overline{R}^2_M = 1 - (1-R^2_M)\frac{n}{n-p_M-1}.$$

The factor $n/(n-p_M-1)$ is the degrees-of-freedom adjustment: adding predictors can reduce RSS, but it also uses up residual degrees of freedom. The full model F has K predictors, so its corresponding denominator is $n-K-1$.

What the identity is comparing. The adjusted R^2 formula expresses the residual sum of squares per residual degree of freedom, after dividing by the same total sum of squares. When we take a ratio for model M and the full model, the common total sum of squares and the common factor n cancel.

Step 1: Write adjusted R^2 for the candidate model

By definition,

$$\overline{R}^2_M = 1 - (1-R^2_M)\frac{n}{n-p_M-1}.$$

Move the adjusted R^2 term to the other side:

$$1-\overline{R}^2_M = (1-R^2_M)\frac{n}{n-p_M-1}.$$

Step 2: Replace $1-R^2_M$ by the RSS ratio

The ordinary coefficient of determination is

$$R^2_M = 1 - \frac{\text{RSS}_M}{SS_Y}.$$

Therefore $1-R^2_M = \text{RSS}_M/SS_Y$, and so

$$1-\overline{R}^2_M = \frac{\text{RSS}_M}{SS_Y}\frac{n}{n-p_M-1}.$$

Rearranging gives

$$\boxed{\frac{\text{RSS}_M}{SS_Y} = \frac{n-p_M-1}{n}(1-\overline{R}^2_M).}$$

Why this step leads to the next step

This is the needed expression for the candidate model. We now write the same identity for the full model, because the desired result is a ratio of the two RSS values.

Step 3: Write the corresponding expression for the full model

The full model contains all K predictors. Hence

$$\overline{R}^2_F = 1 - (1 - R^2_F)\frac{n}{n-K-1}.$$

Using $1 - R^2_F = \mathrm{RSS}_F / SS_Y$ gives

$$\boxed{\frac{\mathrm{RSS}_F}{SS_Y} = \frac{n-K-1}{n}(1-\overline{R}^2_F).}$$

Step 4: Divide the two equations

Divide the expression for RSS_M / SS_Y by the expression for RSS_F / SS_Y:

$$\begin{aligned}\frac{\mathrm{RSS}_M}{\mathrm{RSS}_F} &= \frac{\frac{n-p_M-1}{n}(1-\overline{R}^2_M)}{\frac{n-K-1}{n}(1-\overline{R}^2_F)} \\ &= \frac{n-p_M-1}{n-K-1}\frac{1-\overline{R}^2_M}{1-\overline{R}^2_F}.\end{aligned}$$

This is the algebraically correct identity. The assignment prints the reciprocal degrees-of-freedom factor, $(n-K-1)/(n-p_M-1)$, in this ratio. Unless $p_M = K$, that printed factor is inconsistent with the adjusted-R^2 definition. We therefore retain the question exactly as printed above and report the corrected identity here.

Correction to the printed target identity

The right-hand side printed in the assignment has the degrees-of-freedom factor reversed. The definitions in the supplied lecture notes imply the corrected expression below. In the special case $p_M = K$ (when $M = F$), both forms equal one, but for a proper subset model they are not the same.

Final answer

$$\boxed{\frac{\mathrm{RSS}_M}{\mathrm{RSS}_F} = \frac{(n-p_M-1)(1-\overline{R}^2_M)}{(n-K-1)(1-\overline{R}^2_F)}.}$$

This identity follows directly from the adjusted-R^2 definition, applied once to model M and once to the full model F.

2 Question 2: Expected model error of a reduced predictor

Question (as assigned)

2. Consider the regression model

$$Y_i = \alpha + \gamma_1 x_i + \gamma_2 z_i + \epsilon_i, \qquad i = 1, \ldots, n, \qquad (*)$$

where

$$\sum_i x_i = \sum_i z_i = 0, \qquad \sum_i x_i^2 = \sum_i z_i^2 = 1.$$

Suppose that we want to predict the response Y corresponding to a vector (x, z) using a predictor of the form

$$\widehat{Y} = \bar{Y} + \widetilde{\gamma}_1 x,$$

where $\widetilde{\gamma}_1$ is the least-squares estimate obtained by fitting the model

$$Y_i = \alpha + \gamma_1 x_i + \epsilon_i, \qquad i = 1, \ldots, n.$$

(a) Assuming that model (*) is correct, show that the expected model error

$$\mathbb{E}[ME] = \mathbb{E}\left[(\widehat{Y} - \alpha - \gamma_1 x - \gamma_2 z)^2\right]$$

of the predictor $\widehat{Y}$ is given by

$$\mathbb{E}[ME] = \sigma^2(n^{-1} + x^2) + \gamma_2^2(rx - z)^2, \qquad r = \sum_i x_i z_i.$$

(b) Under what circumstances will $\widehat{Y}$ be a better predictor than the predictor based on both x and z?

Required theory from the supplied lecture notes

The model-selection notes distinguish prediction error from model error. For a fixed future covariate point, the model error is the squared difference between the fitted predictor and the true conditional mean. Its expectation over the training response separates into variance and squared bias:

$$\mathbb{E}\{(\widehat{m} - m)^2\} = \mathrm{Var}(\widehat{m}) + \{\mathbb{E}(\widehat{m}) - m\}^2.$$

This is the scalar bias–variance decomposition.
For a centered regressor, $\sum_i x_i = 0$, the least-squares slope in a simple regression with an intercept is

$$\widetilde{\gamma}_1 = \frac{\sum_i x_i(Y_i - \bar{Y})}{\sum_i x_i^2}.$$

If $\sum_i x_i^2 = 1$, this simplifies to $\widetilde{\gamma}_1 = \sum_i x_i Y_i$.
For centered regressors x, z with unit sums of squares and cross-product $r = \sum_i x_i z_i$, their slope cross-product matrix is

$$G = \begin{pmatrix} 1 & r \\ r & 1 \end{pmatrix}, \qquad G^{-1} = \frac{1}{1 - r^2}\begin{pmatrix} 1 & -r \\ -r & 1 \end{pmatrix},$$

provided $|r| < 1$. Under spherical errors, the covariance of the full-model OLS slope vector is $\sigma^2 G^{-1}$.

A condition to keep in mind. The full model containing both x and z has separately identifiable slope coefficients when $|r| < 1$. This is the non-collinearity condition for the two centered, unit-length predictor vectors. The comparison in part (b) is made under this condition.

Part (a): Derive the expected model error

Step 1: Write the true model in vector form

The true model is

$$Y_i = \alpha + \gamma_1 x_i + \gamma_2 z_i + \epsilon_i,$$

with $\mathbb{E}(\epsilon_i) = 0$, $\mathrm{Var}(\epsilon_i) = \sigma^2$, and spherical independent errors as in the normal linear regression setup. The predictor deliberately fits only an intercept and x:

$$\widehat{Y} = \bar{Y} + \widetilde{\gamma}_1 x.$$

We first derive the behavior of $\bar{Y}$ and $\widetilde{\gamma}_1$ under the true two-predictor model.

Step 2: Find the fitted intercept

Average the true model over i:

$$\bar{Y} = \alpha + \gamma_1 \bar{x} + \gamma_2 \bar{z} + \bar{\epsilon}.$$

The centering assumptions give $\bar{x} = 0$ and $\bar{z} = 0$. Hence

$$\boxed{\bar{Y} = \alpha + \bar{\epsilon}.}$$

Therefore

$$\mathbb{E}(\bar{Y}) = \alpha, \qquad \mathrm{Var}(\bar{Y}) = \mathrm{Var}\left(\frac{1}{n}\sum_i \epsilon_i\right) = \frac{\sigma^2}{n}.$$

Step 3: Find the simple-regression slope estimator

The simple-regression slope is

$$\widetilde{\gamma}_1 = \frac{\sum_i x_i (Y_i - \bar{Y})}{\sum_i x_i^2}.$$

Since $\sum_i x_i = 0$ and $\sum_i x_i^2 = 1$,

$$\widetilde{\gamma}_1 = \sum_i x_i Y_i.$$

Substitute the true model:

$$\begin{aligned}
\widetilde{\gamma}_1 &= \sum_i x_i (\alpha + \gamma_1 x_i + \gamma_2 z_i + \epsilon_i) \\
&= \alpha \sum_i x_i + \gamma_1 \sum_i x_i^2 + \gamma_2 \sum_i x_i z_i + \sum_i x_i \epsilon_i \\
&= \gamma_1 + \gamma_2 r + \sum_i x_i \epsilon_i.
\end{aligned}$$

Thus

$$\boxed{\mathbb{E}(\widetilde{\gamma}_1) = \gamma_1 + \gamma_2 r, \qquad \mathrm{Var}(\widetilde{\gamma}_1) = \sigma^2 \sum_i x_i^2 = \sigma^2.}$$

The omitted predictor z contributes the term $\gamma_2 r$ to the mean of the fitted slope.

Step 4: Establish the covariance between the fitted intercept and slope

We have

$$\mathrm{Cov}(\bar{Y}, \widetilde{\gamma}_1) = \mathrm{Cov}\left(\frac{1}{n}\sum_i \epsilon_i, \sum_j x_j \epsilon_j\right) = \frac{\sigma^2}{n}\sum_j x_j = 0.$$

So the random error in the sample mean and the random error in the simple slope are uncorrelated. Under normality they are also independent, although uncorrelatedness is sufficient for the variance calculation below.

Step 5: Write the prediction error and separate bias from random error

The true conditional mean at the new covariate point (x, z) is $\alpha + \gamma_1 x + \gamma_2 z$. Therefore

$$\begin{aligned}\widehat{Y} - (\alpha + \gamma_1 x + \gamma_2 z) &= \bar{Y} + \widetilde{\gamma}_1 x - \alpha - \gamma_1 x - \gamma_2 z \\ &= (\bar{Y} - \alpha) + x(\widetilde{\gamma}_1 - \gamma_1) - \gamma_2 z.\end{aligned}$$

Use $\bar{Y} - \alpha = \bar{\epsilon}$ and $\widetilde{\gamma}_1 - \gamma_1 = \gamma_2 r + \sum_i x_i \epsilon_i$:

$$\widehat{Y} - (\alpha + \gamma_1 x + \gamma_2 z) = \underbrace{\bar{\epsilon} + x \sum_i x_i \epsilon_i}_{\text{zero-mean random part}} + \underbrace{\gamma_2 (rx - z)}_{\text{bias}}.$$

Step 6: Compute the variance of the random part

The two random terms in the braces have covariance zero by Step 4. Hence

$$\begin{aligned}\operatorname{Var}\left(\bar{\epsilon} + x \sum_i x_i \epsilon_i\right) &= \operatorname{Var}(\bar{\epsilon}) + x^2 \operatorname{Var}\left(\sum_i x_i \epsilon_i\right) \\ &\quad + 2x \operatorname{Cov}\left(\bar{\epsilon}, \sum_i x_i \epsilon_i\right) \\ &= \frac{\sigma^2}{n} + x^2 \sigma^2 + 0 \\ &= \sigma^2 (n^{-1} + x^2).\end{aligned}$$

Step 7: Apply the bias–variance decomposition

If a random variable A has mean b and variance v, then $\mathbb{E}(A^2) = v + b^2$. Here the bias is $\gamma_2(rx - z)$, so

$$\begin{aligned}\mathbb{E}[ME] &= \operatorname{Var}\left(\widehat{Y} - (\alpha + \gamma_1 x + \gamma_2 z)\right) \\ &\quad + \left\{\mathbb{E}(\widehat{Y}) - (\alpha + \gamma_1 x + \gamma_2 z)\right\}^2 \\ &= \sigma^2 (n^{-1} + x^2) + \gamma_2^2 (rx - z)^2.\end{aligned}$$

Final answer

$$\boxed{\mathbb{E}[ME] = \sigma^2 (n^{-1} + x^2) + \gamma_2^2 (rx - z)^2, \qquad r = \sum_{i=1}^{n} x_i z_i.}$$

The first term is prediction variance from estimating the intercept and the x slope. The second term is squared bias caused by omitting z.

Part (b): When is the reduced predictor better than the full predictor?

Meaning of "better". We compare the expected model errors at the same target point (x, z). The reduced predictor is better when its expected squared error is strictly smaller than that of the predictor obtained by fitting both x and z.

Step 1: Find the prediction error of the full two-predictor model

Let $\widehat{\gamma} = (\widehat{\gamma}_1, \widehat{\gamma}_2)^T$ be the full-model OLS slope estimator. Since both regressors are centered, the fitted intercept is $\bar{Y}$. The full-model predictor is

$$\widehat{Y}_F = \bar{Y} + x\widehat{\gamma}_1 + z\widehat{\gamma}_2.$$

The full model is correctly specified, so this predictor is unbiased for $\alpha + \gamma_1 x + \gamma_2 z$. Its model error is therefore just its variance.

Step 2: Compute the covariance matrix of the full slope estimator

The centered slope cross-product matrix is

$$G = \begin{pmatrix} 1 & r \\ r & 1 \end{pmatrix}.$$

Its inverse is

$$G^{-1} = \frac{1}{1-r^2} \begin{pmatrix} 1 & -r \\ -r & 1 \end{pmatrix}.$$

Thus

$$\mathrm{Var}(\widehat{\gamma}) = \frac{\sigma^2}{1-r^2} \begin{pmatrix} 1 & -r \\ -r & 1 \end{pmatrix}.$$

Also, the intercept estimator $\bar{Y}$ has variance σ^2/n and is uncorrelated with both centered slope estimators. Consequently,

$$\begin{aligned} \mathbb{E}[ME_F] &= \mathrm{Var}(\widehat{Y}_F) \\ &= \frac{\sigma^2}{n} + \sigma^2 \begin{pmatrix} x & z \end{pmatrix} G^{-1} \begin{pmatrix} x \\ z \end{pmatrix} \\ &= \frac{\sigma^2}{n} + \frac{\sigma^2}{1-r^2}(x^2 - 2rxz + z^2). \end{aligned}$$

Step 3: Rewrite the full-model variance in terms of $z - rx$

The following identity is useful:

$$x^2 - 2rxz + z^2 = (1-r^2)x^2 + (z - rx)^2.$$

Therefore

$$\mathbb{E}[ME_F] = \frac{\sigma^2}{n} + \sigma^2 x^2 + \frac{\sigma^2}{1-r^2}(z - rx)^2.$$

Step 4: Subtract the full-model error from the reduced-model error

From part (a),

$$\mathbb{E}[ME_R] = \frac{\sigma^2}{n} + \sigma^2 x^2 + \gamma_2^2 (rx - z)^2.$$

Since $(rx - z)^2 = (z - rx)^2$,

$$\begin{aligned} \mathbb{E}[ME_R] - \mathbb{E}[ME_F] &= \gamma_2^2 (z - rx)^2 - \frac{\sigma^2}{1-r^2}(z - rx)^2 \\ &= (z - rx)^2 \left(\gamma_2^2 - \frac{\sigma^2}{1-r^2} \right). \end{aligned}$$

Step 5: Read off the circumstances for the reduced predictor to be better

The reduced predictor is strictly better precisely when

$$\mathbb{E}[ME_R] - \mathbb{E}[ME_F] < 0.$$

There are two factors to consider:

- If $z = rx$, then $(z - rx)^2 = 0$ and the two predictors have equal expected model error at this particular target point, regardless of γ_2.
- If $z \neq rx$, the sign is determined by the second factor. The reduced predictor is better exactly when

$$\gamma_2^2 < \frac{\sigma^2}{1 - r^2}, \quad \text{equivalently} \quad |\gamma_2| < \frac{\sigma}{\sqrt{1 - r^2}}.$$

If equality holds in the threshold, the two predictors tie for every target point. If $\gamma_2^2 > \sigma^2/(1 - r^2)$, the full predictor is better whenever $z \neq rx$.

Final answer

Assuming $|r| < 1$ so that the full two-predictor model is estimable,

$$\boxed{\mathbb{E}[ME_R] - \mathbb{E}[ME_F] = (z - rx)^2 \left(\gamma_2^2 - \frac{\sigma^2}{1 - r^2}\right).}$$

Thus, for a target point with $z \neq rx$, the reduced predictor $\bar{Y} + \widetilde{\gamma}_1 x$ is better precisely when

$$\boxed{|\gamma_2| < \frac{\sigma}{\sqrt{1 - r^2}}.}$$

If $z = rx$, both predictors have the same expected model error at that point. The condition expresses a bias–variance trade-off: omitting z is beneficial when the squared omission bias is smaller than the variance saved by not estimating its coefficient.

End of Problems Set 4 solutions